Question

Compound XYZ_3 is a common substance in life, with mineral distribution in nature. Its crystal structure is similar to that of $NaCl$ crystals in terms of stacking, but the selection method of its proper unit cell is different from that of $NaCl$ crystals. If it is stacked in the same way as $NaCl$, the volume of the proper unit cell in this case is 2/3 of its actual proper unit cell. In XYZ_3, X is coordinated by 6 Z at equal distances; ${YZ_3}^{n-}$ is parallel to the ab plane.

If the proper unit cell of XYZ_3 has X^{n+} as a vertex, write down the atomic coordinates of all Y in the unit cell.

A. $\left(\frac{2}{3},\frac{1}{3},\frac{1}{12}\right)$, $\left(0,0,\frac{1}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{5}{12}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{7}{12}\right)$, $\left(0,0,\frac{3}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{11}{12}\right)$

or

$\left(\frac{2}{3},\frac{1}{3},\frac{11}{12}\right)$, $\left(0,0,\frac{3}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{7}{12}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{5}{12}\right)$, $\left(0,0,\frac{1}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{1}{12}\right)$

B. $(0,0,0)$, $\left(\frac{1}{3},\frac{2}{3},\frac{1}{6}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{1}{3}\right)$, $\left(0,0,\frac{1}{2}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{2}{3}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{5}{6}\right)$

C. $(0,0,0)$, $\left(\frac{1}{3},\frac{2}{3},\frac{5}{6}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{2}{3}\right)$, $\left(0,0,\frac{1}{2}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{1}{3}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{1}{6}\right)$

D. $\left(\frac{3}{4},\frac{3}{4},0\right)$, $\left(\frac{3}{4},0,\frac{3}{4}\right)$, $\left(0,\frac{3}{4},\frac{3}{4}\right)$, $\left(\frac{3}{4},\frac{3}{4},\frac{3}{4}\right)$

E. $(0,0,0)$, $\left(\frac{1}{3},\frac{2}{3},\frac{1}{6}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{1}{3}\right)$, $\left(0,0,\frac{1}{2}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{2}{3}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{5}{6}\right)$

F. $\left(\frac{2}{3},\frac{1}{3},\frac{1}{12}\right)$, $\left(0,0,\frac{1}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{5}{12}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{7}{12}\right)$, $\left(0,0,\frac{3}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{11}{12}\right)$

G. $\left(\frac{2}{3},\frac{1}{3},\frac{11}{12}\right)$, $\left(0,0,\frac{3}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{7}{12}\right)$, $\left(\frac{2}{3},\frac{1}{3},\frac{5}{12}\right)$, $\left(0,0,\frac{1}{4}\right)$, $\left(\frac{1}{3},\frac{2}{3},\frac{1}{12}\right)$

Answer

Correct Answer: A

Detailed Explanation

The typical characteristic of the NaCl structure is that the two types of ions are stacked in a ccp arrangement, so the same stacking form should also exist in $\mathrm{XYZ_3}$.

CHECKPOINT 1 PTS

$\mathrm{XYZ_3}$ has ccp stacking

The primitive unit cell of NaCl belongs to the cubic crystal system and contains 4 Na and 4 Cl. One cycle of its stacking mode, namely the AcBaA type stacking (containing 2 Na and 2 Cl), has a unit cell volume that is 0.5 times the volume of the NaCl unit cell. Therefore, for the primitive unit cell of $\mathrm{XYZ_3}$, if the volume is to satisfy 1.5 times the volume of the primitive unit cell of the NaCl stacking form (if it is stacked in the same stacking method as NaCl, the primitive unit cell volume in this case is 2/3 of its actual primitive unit cell), then 3 AcBaA-sized unit cells need to overlap. Combined with symmetry requirements, it should be AbC'aBcA'bCaB'c.

CHECKPOINT 1 PTS

AbC'aBcA'bCaB'c

The Y atom itself has no orientation and fills the octahedral voids of the X atoms.

CHECKPOINT 1 PTS

Y atom fills the octahedral voids of the X atoms

And according to the different choices of the B and C layer positions, there can be two sets of coordinates.

CHECKPOINT 1 PTS

Two sets of coordinates

Therefore, the coordinates are $\left(\frac{2}{3}, \frac{1}{3}, \frac{1}{12}\right)$, $\left(0, 0, \frac{1}{4}\right)$, $\left(\frac{1}{3}, \frac{2}{3}, \frac{5}{12}\right)$, $\left(\frac{2}{3}, \frac{1}{3}, \frac{7}{12}\right)$, $\left(0, 0, \frac{3}{4}\right)$, $\left(\frac{1}{3}, \frac{2}{3}, \frac{11}{12}\right)$

CHECKPOINT 1 PTS

Coordinates are $\left(\frac{2}{3}, \frac{1}{3}, \frac{1}{12}\right)$, $\left(0, 0, \frac{1}{4}\right)$, $\left(\frac{1}{3}, \frac{2}{3}, \frac{5}{12}\right)$, $\left(\frac{2}{3}, \frac{1}{3}, \frac{7}{12}\right)$, $\left(0, 0, \frac{3}{4}\right)$, $\left(\frac{1}{3}, \frac{2}{3}, \frac{11}{12}\right)$